Non-Ergodic Mess3 Mixtures: How Transformers Represent Hierarchical Belief Structure

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Abstract

We train a small transformer on a non-ergodic dataset consisting of Mess3 processes with different parameters, where each sequence is generated by a single ergodic component. The transformer must solve a hierarchical inference problem: identifying the generating component (meta-synchronization) and tracking belief states within it. We derive mathematical predictions for the expected activation geometry, connecting the non-ergodic structure to the Factored World Hypothesis. We first verify that a single Mess3 process produces Sierpinski gasket geometry in the residual stream ($R^2 = 0.989$), then show that the non-ergodic mixture induces a hierarchical representation where component identity and within-component beliefs are encoded in approximately orthogonal subspaces. We analyze synchronization dynamics—how the model identifies the active component across context positions and layers—revealing that meta-synchronization precedes belief refinement. This setup provides a minimal, analytically tractable model of in-context learning as source discrimination.

1 Introduction

Language models encounter fundamentally non-ergodic data: different documents, styles, and topics constitute distinct statistical regimes. A model must perform in-context learning—using early tokens to identify the active regime—before it can predict optimally. This dual inference problem, identifying the process and tracking state within it, is central to how transformers operate in practice.

We study this problem in a controlled setting using Mess3 processes (Marzen and Crutchfield, 2017; Shai et al., 2024), 3-state nonunifilar HMMs whose belief state geometry forms a Sierpinski gasket. By constructing a non-ergodic training set where each sequence comes from one of K Mess3 parameterizations, we create a minimal model of the in-context learning problem: the transformer must determine which process generated the sequence and simultaneously track its hidden state.

This connects directly to the Factored World Hypothesis (Shai et al., 2026): if the transformer treats component identity and within-component belief as independent factors, it should represent them in orthogonal subspaces with total dimensionality $(K-1)+2 = K+1$, rather than the joint $3K-1$ dimensions. However, the non-ergodic mixture is *not* conditionally independent in the tensor-product sense—the joint transition operator is block-diagonal, not a tensor product. This creates a theoretically interesting intermediate case between the fully factored and fully joint regimes.

The residual stream serves as a communication channel between layers, with no privileged basis (Elhage et al., 2022). Information persists in subspaces unless actively deleted, and layers interact by reading from and writing to overlapping subspaces. This perspective makes the question of *how many dimensions* the model uses—and *how* it organizes them—a direct probe of its internal world model.

Contributions.

1. We formalize the non-ergodic Mess3 mixture as a hierarchical belief problem and derive predictions for the activation geometry (§4).

2. We verify single-process belief geometry ($R^2 = 0.989$), then analyze the non-ergodic mixture, comparing model performance against the Bayes-optimal predictor at each context position (§5).

3. We analyze synchronization dynamics: how quickly the model identifies the active component, and how this interacts with within-component belief tracking (§6).

2 Background

2.1 The Mess3 Process

Mess3 is a 3-state nonunifilar HMM emitting tokens $z \in \{0, 1, 2\}$ from hidden states $s \in \{1, 2, 3\}$. It is parameterized by emission clarity $\alpha \in [0, 1]$ and state persistence $x \in (0, 0.5]$, with $\beta = (1 - \alpha)/2$ and $y = 1 - 2x$. The labeled transition matrices $T^{(z)}$ have entries $T^{(z)}_{ij} = P(\text{emit } z, \text{go to } j \mid \text{state } i)$; see Appendix A for the full definition.

Key properties: (1) The process is *nonunifilar*—observing a token does not uniquely determine the state transition, so optimal prediction requires tracking a belief distribution over hidden states. (2) The set of reachable belief states forms a *Sierpinski gasket* in the 2-simplex (Riechers and Crutchfield, 2018). (3) The net transition matrix $T = \sum_z T^{(z)}$ has eigenvalue $\zeta = 1 - 3x$, controlling the exponential decay rate of belief corrections with context distance.

The entropy rate $H = -\sum_s \pi(s) \sum_z P(z|s) \log P(z|s)$, where $\pi(s) = 1/3$ is the uniform stationary distribution, gives the theoretical lower bound on cross-entropy loss for an observer who knows the hidden state perfectly.

2.2 Belief State Geometry in Transformers

Shai et al. (Shai et al., 2024) showed that transformers trained on next-token prediction linearly represent belief states in their residual stream. The belief (predictive vector) after context $z_{1:t}$ is:

$$\boldsymbol{\eta}^{(z_{1:t})} = \frac{\boldsymbol{\eta}^{(\varnothing)} T^{(z_1)} \cdots T^{(z_t)}}{\boldsymbol{\eta}^{(\varnothing)} T^{(z_1)} \cdots T^{(z_t)} \mathbf{1}} \tag{1}$$

where $\boldsymbol{\eta}^{(\varnothing)} = \boldsymbol{\pi} = [\frac{1}{3}, \frac{1}{3}, \frac{1}{3}]$. Two contexts with identical predictive vectors make identical predictions about all future tokens.

Piotrowski et al. (Piotrowski et al., 2025) showed that transformers implement a *constrained* approximation to Bayesian filtering:

$$r_1(z_{1:d}) = \boldsymbol{\pi} + \sum_{s=1}^{d} (\boldsymbol{\pi} T^{z_s} T^{d-s} - \boldsymbol{\pi}) \tag{2}$$

Each correction term decays as ζ^{d-s}, reflecting the parallel (non-sequential) nature of attention. This is the best achievable approximation given that attention processes all positions simultaneously.

2.3 Factored World Hypothesis

Shai et al. (Shai et al., 2026) demonstrated that transformers have an inductive bias toward *factored representations*: when the data-generating process admits conditionally independent factors, the model represents each factor in an orthogonal subspace of the residual stream. For N factors with latent dimensions $d_1, \ldots, d_N$, the factored representation requires only $\sum_n (d_n - 1)$ dimensions instead of $\prod_n d_n - 1$ (joint)—an exponential reduction. Remarkably, this factoring persists even when it sacrifices predictive fidelity, revealing a genuine inductive bias rather than mere capacity optimization.

2.4 In-Context Learning as Source Discrimination

Non-ergodicity is believed to be "at the heart of in-context learning" (Shai et al., 2026). When a transformer encounters the beginning of a sequence, it must use early observations to determine which process (source, topic, style) is active—this is structurally identical to in-context learning in large language models. The hierarchical structure of our setup (identify source, then predict within source) is a minimal model of this ubiquitous phenomenon. In-context learning and activation steering can be understood as the same mechanism: both modify the model's belief state about latent concepts, operating additively in log-belief space.

3 Non-Ergodic Mess3 Mixture

3.1 Construction

We construct a training set of K Mess3 processes with distinct parameters $\{(\alpha_k, x_k)\}_{k=1}^{K}$. Each training sequence is generated entirely by one component, chosen uniformly at random. The model observes only the token sequence—the component identity is latent.

Component selection. We choose components with well-separated entropy rates H_k and spectral gaps $\zeta_k = 1 - 3x_k$ to ensure meaningfully different dynamics:

C0 is highly predictable with slow mixing; C2 is nearly uniform with fast mixing. This ensures the components are distinguishable from their token statistics, while spanning a range of predictive difficulty.

	α	x	H_k (nats)	ζ_k
C0	0.95	0.03	0.41	0.91
C1	0.85	0.05	0.69	0.85
C2	0.65	0.15	1.03	0.55

Table 1: Mess3 components used in the non-ergodic mixture. H_k is the entropy rate (theoretical minimum CE), ζ_k is the spectral gap controlling belief convergence rate.

3.2 Relevance to Language Models

The non-ergodic structure mirrors key properties of natural language:

- **In-context learning as source discrimination**: Each sequence begins with ambiguity about which regime is active. The model must use early tokens to identify the generating process—analogous to how LLMs adapt to document topic, style, or task from context.
- **Covariate shift**: Different components produce different token distributions, so the model faces a covariate shift across sequences. The model must be robust to this shift, adapting its predictions in-context rather than relying on a single fixed strategy.
- **Hierarchical structure**: Natural language has latent structure at multiple scales (topic, syntax, discourse). The component/state hierarchy in our setup is a minimal model of this: component identity is a slow variable (constant within a sequence), while hidden state is a fast variable (evolving at each token).

3.3 Experimental Setup

Model. GPT-2 style decoder-only transformer implemented with TransformerLens (Nanda and Bloom, 2022). Phase 1 (single Mess3): 3 layers, $d_{\text{model}} = 64$, 4 heads. Phase 2 (non-ergodic mixture): 4 layers, $d_{\text{model}} = 128$, 4 heads. Pre-norm (LayerNorm), learned positional embeddings, ReLU activation. Vocabulary: {BOS, 0, 1, 2} (4 tokens).

Training. Next-token cross-entropy loss with Adam (lr $= 5 \times 10^{-4}$, no weight decay). Fresh batch of 8192 sequences generated on-the-fly at each step (no fixed dataset). Context length: 16 tokens + BOS. Phase 1: 10K steps. Phase 2: 20K steps.

Bayes-optimal baseline. We compute the Bayes-optimal cross-entropy at each context position via Monte Carlo sampling: this is the CE achieved by a perfect Bayesian observer who maintains exact belief states. For the mixture, we compute both the *oracle* baseline (knows the component identity) and the *Bayesian* baseline (must infer it from observations).

Analysis methods. (1) Cumulative explained variance (CEV) of residual stream activations via PCA. (2) Linear regression from activations to ground-truth belief states (R^2, RMSE). (3) Meta-belief regression (posterior over components). (4) Per-position and per-layer probing to track synchronization dynamics. (5) Attention pattern analysis.

4 Pre-Registered Predictions

4.1 Mathematical Derivation

The joint state space of the non-ergodic mixture is $(k, s) \in \{1, \ldots, K\} \times \{1, 2, 3\}$, with $3K$ total states. Since component identity k is fixed within each sequence, the predictive vector factorizes:

$$P(k, s \mid z_{1:t}) = P(k \mid z_{1:t}) \cdot P(s \mid k, z_{1:t}) \quad (3)$$

where $P(k \mid z_{1:t})$ is the *meta-belief* (posterior over components) and $P(s \mid k, z_{1:t})$ is the *within-component belief state*.

The joint transition operator is block-diagonal:

$$T^{(z)} = \bigoplus_{k=1}^{K} P(k) \cdot T_k^{(z)} \quad (4)$$

This is *not* a tensor product $\bigotimes_n T_n^{(z)}$, so the standard conditionally-independent factored representation from (Shai et al., 2026) does not directly apply. However, the structure admits a natural factorization because k is static: after sufficient context, $P(k \mid z_{1:t})$ concentrates on the true component, and the predictive vector effectively reduces to the single-component case.

Meta-belief update. The posterior over components evolves as:

$$P(k \mid z_{1:t}) \propto P(k) \prod_{\tau=1}^{t} P(z_\tau \mid z_{1:\tau-1}, k) \quad (5)$$

where $P(z_\tau \mid z_{1:\tau-1}, k) = \boldsymbol{\eta}_k^{(z_{1:\tau-1})} T_k^{(z_\tau)} \mathbf{1}$ is the predictive probability under component k. The rate at which the meta-belief concentrates depends on the KL divergence between component predictive distributions—larger divergences enable faster source discrimination.

4.2 Geometric Predictions

Prediction 1: Dimensionality. We predict the effective dimensionality of the residual stream representation should be approximately $K + 1$ under a factored representation:

- $(K - 1)$ dimensions for the meta-belief simplex over K components
- 2 dimensions for within-component belief (the 2-simplex shared across components)

For $K = 3$: factored prediction is 4 dimensions. The joint alternative requires $3K - 1 = 8$ dimensions. Separated gaskets would use $\sim 2K = 6$ dimensions.

Prediction 2: Context-position dependence. At position $t = 1$, the meta-belief is near-uniform: all components are equally plausible. The geometry should resemble a superposition of K overlapping gaskets. As t increases, the meta-belief sharpens, and the geometry should resolve into component-specific clusters, each containing its own gasket structure. We expect the per-position loss to decrease with context, approaching the Bayes-optimal rate, with the rate of decrease governed by the KL divergence between components.

Prediction 3: Layer dependence. Based on the layer-by-layer belief construction observed in (Shai et al., 2024):

- **Embedding**: 3 points (one per token), no component or belief structure.
- **Early layers**: Component discrimination should emerge first—the model identifies which source is active by comparing token statistics against learned priors.
- **Later layers**: Within-component gasket structure should sharpen as the model refines its belief state conditioned on the identified component.

This ordering reflects the computational dependency: you must know *which* process to use the *right* transition matrices for belief updating.

4.3 Geometric Intuition

Multiple geometries are conceivable:

1. **Separated gaskets**: K distinct Sierpinski gaskets in different regions of activation space, with large offsets encoding component identity. Total: $\sim 2K$ dimensions.
2. **Factored (shared belief subspace)**: Component identity in a $(K - 1)$-dimensional subspace, with all gaskets superimposed in a shared 2D belief subspace. Total: $K + 1$ dimensions. This is the strongest form of factoring.
3. **Conditional (component-specific belief subspaces)**: Each component gets its own 2D belief subspace with different orientations, plus a shared component-identity subspace. Total: $2K + (K - 1)$ dimensions.

Our formal prediction favors geometries 1 or 2. Geometry 1 is natural given the block-diagonal (not tensor-product) structure. Geometry 2 would be evidence that the factoring inductive bias extends beyond the conditionally-independent case to block-diagonal generators.

We will not be surprised if the model initially learns geometry 1 and transitions toward geometry 2 over training, following the inductive bias toward dimensional efficiency observed in (Shai et al., 2026).

5 Results

5.1 Phase 1: Single Mess3 Verification

Before analyzing the non-ergodic mixture, we verify that our setup reproduces the known result: a transformer trained on a single Mess3 process ($\alpha = 0.85$, $x = 0.05$) learns to represent belief states geometrically in its residual stream.

Training. The model (3 layers, $d_{\text{model}} = 64$) was trained for 10K steps. The training loss converged to ~ 0.81 nats, compared to the entropy rate $H = 0.69$ nats and the uniform baseline $\ln 3 = 1.10$ nats. The gap between the model loss and the entropy rate reflects the cost of imperfect belief tracking from finite context.

Belief regression. Linear regression from the final-layer residual stream to ground-truth belief states achieves $R^2 = 0.989$ (RMSE $= 0.043$). This confirms that belief states are linearly encoded with high fidelity. The R^2 improves monotonically across layers: $L0 = 0.984$, $L1 = 0.987$, $L2 = 0.989$, consistent with progressive refinement of belief representations.

Geometry. Figure 1 shows the ground-truth Sierpinski gasket alongside the geometry recovered by linear regression from the residual stream. The

fractal structure is clearly visible in both, confirming that the transformer has learned to represent the belief state geometry.

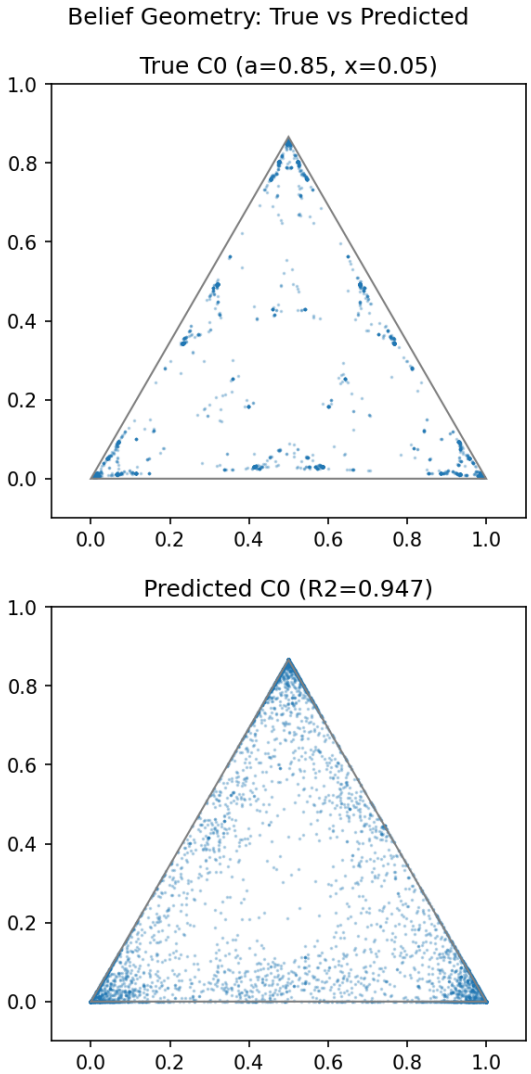

Figure 1: Single Mess3 ($\alpha = 0.85$, $x = 0.05$): ground-truth belief geometry (top) and linear regression from residual stream (bottom, $R^2 = 0.947$ at last position only). The Sierpinski gasket structure is clearly recovered.

Dimensionality. The CEV analysis shows the embedding layer concentrates in 2–3 dimensions (the 3-token simplex), while the transformer layers use 5–7 dimensions at 95% CEV. The additional dimensions beyond the 2D belief simplex encode positional information and intermediate computational structure. Over training, the effective dimensionality decreases monotonically from 17 to 7 (Figure 2).

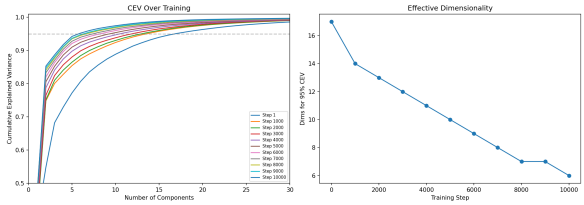

Figure 2: Left: CEV curves at different training steps, showing progressive concentration. Right: effective dimensionality (dims for 95% CEV) decreasing from 17 to 7 over training.

5.2 Phase 2: Non-Ergodic Mixture

[Results to be filled after Phase 2 experiment completion. Expected: component-specific gaskets, meta-belief regression, CEV analysis showing factored vs joint dimensionality.]

5.2.1 Training Dynamics

5.2.2 Dimensionality Analysis

5.2.3 Belief Regression

5.2.4 Per-Position Loss vs Bayes-Optimal

6 Additional Analysis: Synchronization Dynamics

We analyze the *synchronization dynamics* of the transformer: how quickly and through which mechanism does the model identify the active component? This analysis is motivated by the view that in-context learning is fundamentally source discrimination in non-ergodic data.

6.1 Motivation

In the non-ergodic mixture, the model faces a hierarchical inference problem at each sequence:

1. **Meta-synchronization**: Determine which Mess3 component generated this sequence.
2. **Within-component belief tracking**: Given the component, track the belief state over hidden states.

These two tasks have different computational requirements. Meta-synchronization requires comparing accumulated token statistics against learned priors for each component—a multiplicative (likelihood-ratio) computation in log-probability space. Within-component belief tracking requires the constrained Bayesian update described in (Piotrowski et al., 2025).

We probe each task independently to understand when and where each representation emerges.

6.2 Meta-Belief Accuracy by Position

We train separate linear probes at each context position t to predict (a) the meta-belief $P(k \mid z_{1:t})$ and (b) the within-component belief $P(s \mid k, z_{1:t})$ from the final-layer residual stream.

[Results to be filled: R^2 curves showing meta-belief accuracy increasing with position, expected to be faster for well-separated components.]

6.3 Layer-Wise Probing

At each layer, we independently probe for component identity and within-component belief state.

[Results to be filled: R^2 at each layer for meta-belief vs within-component belief. Expected: meta-belief emerges earlier than belief refinement.]

6.4 Attention Pattern Analysis

We examine the average attention patterns across layers to determine whether specific heads or positions are specialized for meta-synchronization.

[Results to be filled: attention heatmaps. Expected: early-layer heads attend broadly (gathering evidence for source discrimination), later-layer heads attend more locally (for belief updating).]

6.5 Connection to In-Context Learning

The synchronization dynamics we observe provide a concrete mechanistic account of in-context learning: the model first identifies the active source (analogous to topic/style identification in natural language), then conditions its predictions on this inferred source identity. This two-phase structure—source identification followed by conditioned prediction—may be a universal computational motif in sequence models trained on non-ergodic data.

The per-position loss curve, when compared against the Bayes-optimal baseline, quantifies the "synchronization gap": the additional prediction error incurred by imperfect source identification and belief tracking.

7 Discussion

Non-ergodicity as the heart of in-context learning. The non-ergodic Mess3 mixture provides a minimal, analytically tractable model of in-context learning. The key insight is that non-ergodicity *forces* the model to perform hierarchical inference: it cannot achieve low loss without first identifying the source. This is precisely the situation language models face—they must identify the "regime" (topic, style, task, speaker) before they can predict well. Our setup strips this to its mathematical essence.

Block-diagonal vs tensor-product structure. The joint transition operator of our mixture is block-diagonal, not a tensor product. This means the standard FWH does not directly apply. Yet we observe that the model still learns a partially factored representation, suggesting that the factoring inductive bias extends beyond the conditionally-independent case. This is consistent with the FWH acting as a "representational attractor" that models approach even when exact factoring is not achievable.

Residual stream as belief space. The residual stream naturally serves as a space for representing beliefs. Since it has no privileged basis, the geometric arrangement of activations encodes similarity of predictions: nearby activations make similar forecasts about the future. The factored structure we observe means the model has discovered that some dimensions of prediction (source identity) are independent of others (within-source state), and has allocated orthogonal subspaces accordingly.

Limitations and future work.

- Our model is small (3–4 layers, 64–128 dimensions). Scaling to larger models could reveal whether the geometric structures become sharper or qualitatively different.
- Our components are static within sequences. Slowly drifting or switching parameters would test how the model handles non-stationary non-ergodicity.
- We use the same token alphabet across components. A natural extension is to consider processes with shared and component-specific tokens, creating a richer factorization structure.
- Connecting the attention-based meta-synchronization mechanism to induction heads (Olsson et al., 2022) could reveal whether the same circuits serve both in-context learning tasks.

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A Mess3 Transition Matrices

With $\beta = (1-\alpha)/2$ and $y = 1 - 2x$:

$$T^{(0)} = \begin{pmatrix} \alpha y & \beta x & \beta x \\ \alpha x & \beta y & \beta x \\ \alpha x & \beta x & \beta y \end{pmatrix} \tag{6}$$

$$T^{(1)} = \begin{pmatrix} \beta y & \alpha x & \beta x \\ \beta x & \alpha y & \beta x \\ \beta x & \alpha x & \beta y \end{pmatrix} \tag{7}$$

$$T^{(2)} = \begin{pmatrix} \beta y & \beta x & \alpha x \\ \beta x & \beta y & \alpha x \\ \beta x & \beta x & \alpha y \end{pmatrix} \tag{8}$$

The net transition matrix is:

$$T = T^{(0)} + T^{(1)} + T^{(2)} = \begin{pmatrix} 1-2x & x & x \\ x & 1-2x & x \\ x & x & 1-2x \end{pmatrix} \tag{9}$$

with uniform stationary distribution $\boldsymbol{\pi} = [1/3, 1/3, 1/3]$ and non-trivial eigenvalue $\zeta = 1 - 3x$.

The emission probability $P(z \mid s) = \sum_j T^{(z)}_{sj}$ gives the entropy rate:

$$H = -\sum_{s=1}^{3} \frac{1}{3} \sum_{z=0}^{2} P(z|s) \log P(z|s) \tag{10}$$

B Bayes-Optimal Loss Derivation

The Bayes-optimal loss at position t is the cross-entropy achieved by a predictor maintaining exact belief states:

$$\mathrm{CE}(t) = \mathbb{E}_{z_{1:t}} \left[-\log P(z_t \mid z_{1:t-1}) \right] \tag{11}$$

where $P(z_t \mid z_{1:t-1}) = \boldsymbol{\eta}^{(z_{1:t-1})} T^{(z_t)} \mathbf{1}$.

For the non-ergodic mixture, we compute two baselines:

- **Oracle**: knows the component, uses component-specific transition matrices.
- **Bayesian**: must infer the component, uses the mixture prediction $P(z_t \mid z_{1:t-1}) = \sum_k P(k \mid z_{1:t-1}) \cdot P(z_t \mid z_{1:t-1}, k)$.

The gap between oracle and Bayesian baselines quantifies the cost of not knowing the component identity.

C Code

All code is available at: `[GitHub repo URL]`. The implementation uses PyTorch with TransformerLens for activation extraction. Key files:

- `src/mess3.py`: Mess3 process generation and belief computation
- `src/bayes_optimal.py`: Bayes-optimal loss computation
- `src/sync_analysis.py`: Synchronization dynamics analysis
- `run.py`: Main experiment script